Q1:
The execution scenario tested was conducted by running: **java q1.java 500 600 t 12** where t = the number of threads being tested for each of the scenarios. The results of the plot file were extracted by averaging 10 data points for each of the execution scenaris, and calculating the speedup with the formula: Speedup = (Average Time for a Single-Thread) / (Average Time for Multi-Threading). This resulted in an average execution time shown in the following table:

Number of threads	Average Execution Time (ms)	Speedup
1	9	1
2	6	1.5
3	5.6	1.60714285714286
4	5.3	1.69811320754717
6	5.5	1.636363636364
12	5.4	1.666666666667

The test machine I am using is a 16 core computer. As separate cores on a machine are able handle multiple tasks simultaneously, this allows the concurrency of multithreading to be more efficient.
The synchronization strategy revolved around ensuring that the circles arraylist is always synchronized such that after generating random circles, all changes to the circles array is visible when checking for overlap and if the snowman is within the bounds of the image as it is contained within a synchronized block, ensuring only one thread can check for overlap at a time in this critical section. This allows the threads to individually generate random locations to fit the snowman and ensures that as the circles array is synchronized, no 2 threads would choose the same location and check for overlap at the same time such that the location is not contained within the circles array. This guarantees mutual exclusion and thus prevents data races. As it can be seen, the speedup stops increasing after 4 concurrent threads are used, as execution time can still be high due to some of the threads choosing random locations that contain overlap, forcing the do while loop to regenerate another location, increasing the execution time. As the average execution time at 4+ threads is already very small, each repeated generation would cause a noticeable increase in execution time, affecting the average execution time.

Q2.
PS C:\Users\xiayi\Desktop\COMP-409\a1> java SnakesAndLadders.java 300 100 5
Sample Output:
00000000 Adder snake 41 60
00000011 Adder snake 80 93
00000011 Adder snake 49 52
00000011 Adder snake 27 67
00000011 Adder snake 31 75
00000011 Adder snake 13 44
00000011 Adder snake 64 85
00000011 Adder snake 45 97
00000011 Adder snake 1 82
00000011 Adder snake 53 88
00000011 Adder ladder 84 29
00000011 Adder ladder 69 38
00000011 Adder ladder 43 21
00000011 Adder ladder 55 46
00000011 Adder ladder 40 9
00000011 Adder ladder 56 37
00000011 Adder ladder 89 30
00000011 Adder ladder 90 86
00000011 Adder ladder 94 34
00000011 Adder snake 48 91
00000011 Remover ladder 90 86
00000011 Player 1
00000039 Player 4
00000063 Player 8
00000111 Player 12
00000124 Remover snake 27 67
00000135 Player 13
00000159 Player 19
00000198 Player 25
00000226 Remover ladder 56 37
00000238 Player 27
00000263 Player 30
00000263 Player 30 89
00000289 Player 93
00000289 Player 93 80
00000326 Adder snake 11 58
00000326 Player 86
00000327 Remover ladder 69 38
00000367 Player 91
00000367 Player 91 48
00000417 Player 53

00000427 Remover ladder 94 34
00000455 Player 58
00000455 Player 58 11
00000492 Player 16
00000528 Remover ladder 43 21
00000538 Player 17
00000587 Player 19
00000618 Player 23
00000634 Remover ladder 84 29
00000634 Adder snake 3 71
00000650 Player 27
00000695 Player 30
00000695 Player 30 89
00000742 Remover snake 31 75
00000744 Player 93
00000744 Player 93 80
00000782 Player 81
00000818 Player 82
00000818 Player 82 1
00000843 Remover snake 49 52
00000852 Player 3
00000880 Player 9
00000880 Player 9 40
00000907 Player 44
00000907 Player 44 13
00000942 Adder snake 42 62
00000944 Remover snake 41 60
00000952 Player 19
00000985 Player 22
00001027 Player 26
00001044 Remover ladder 89 30
00001076 Player 32
00001112 Player 36
00001145 Remover ladder 55 46
00001150 Player 41
00001175 Player 43
00001213 Player 45
00001245 Remover ladder 40 9
00001245 Adder snake 10 26
00001252 Player 51
00001296 Player 55
00001330 Player 60
00001346 Remover snake 11 58
00001374 Player 65

00001403 Player 70
00001429 Player 72
00001446 Remover snake 64 85
00001451 Player 78
00001496 Player 79
00001521 Player 82
00001521 Player 82 1
00001546 Adder ladder 57 40
00001547 Player 4
00001547 Remover ladder 57 40
00001592 Player 10
00001630 Player 16
00001647 Remover snake 80 93
00001674 Player 22
00001711 Player 26
00001711 Player 26 10
00001747 Remover snake 13 44
00001752 Player 16
00001797 Player 19
00001834 Player 21
00001846 Adder ladder 96 32
00001847 Remover snake 3 71
00001861 Player 25
00001903 Player 28
00001933 Player 30
00001948 Remover ladder 96 32
00001970 Player 31
00002015 Player 37
00002050 Remover snake 10 26
00002058 Player 43
00002087 Player 44
00002114 Player 50
00002147 Adder ladder 34 3
00002149 Player 52
00002150 Remover snake 45 97
00002187 Player 54
00002213 Player 57
00002249 Player 62
00002249 Player 62 42
00002251 Remover ladder 34 3
00002288 Player 47
00002332 Player 53
00002351 Remover snake 1 82
00002378 Player 58

00002403 Player 63
00002429 Player 65
00002449 Adder snake 30 46
00002458 Player 70
00002483 Player 72
00002533 Player 73
00002552 Remover snake 42 62
00002577 Player 77
00002615 Player 79
00002658 Player 82
00002692 Player 84
00002725 Player 85
00002749 Adder ladder 41 15
00002755 Remover snake 30 46
00002759 Player 88
00002759 Player 88 53
00002792 Player 57
00002815 Player 62
00002863 Remover snake 48 91
00002901 Player 74
00002948 Player 77
00002964 Remover ladder 41 15
00002972 Player 81
00003006 Player 82
00003039 Player 85
00003050 Adder snake 18 83
00003064 Remover snake 18 83
00003082 Player 90
00003128 Player 91
00003175 Player 93
00003217 Player 95
00003266 Player 96
00003289 Player wins
00003353 Adder ladder 74 40
00003400 Remover snake 53 88
00003450 Player 1
00003498 Player 5
00003500 Remover ladder 74 40
00003533 Player 10
00003573 Player 11
00003619 Player 15
00003666 Player 18
00003666 Player 18 43
00003714 Remover ladder 43 18

00003730 Player 49
00003779 Player 54
00003829 Player 57
00003878 Player 59
00003905 Player 61
00003930 Player 62
00003969 Player 67
00003981 Adder ladder 19 4
00004007 Player 71
00004026 Remover ladder 19 4
00004044 Player 76
00004087 Player 77
00004137 Player 81
00004162 Player 85
00004197 Player 89
00004244 Player 94
00004268 Player 96
00004281 Adder snake 35 50
00004314 Player wins
00004330 Remover snake 35 50
00004474 Player 1
00004515 Player 6
00004541 Player 10
00004572 Player 16
00004582 Adder ladder 18 7
00004604 Player 20
00004652 Remover ladder 18 7
00004684 Player 26
00004727 Player 31
00004771 Player 37
00004799 Player 41
00004822 Player 42
00004872 Player 45
00004888 Adder snake 14 37
00004936 Player 48
00004984 Player 53
00005009 Player 58